CO 250 Review 4: Simplex set up

Goal: Setting up simplex. ① Standard equality form. ② Basis. ③ Canonical form.

Standard equality form (SEF)

$$\begin{aligned} \max\ & c^T x \\ \text{s.t. } & Ax = b \\ & x \geq 0 \end{aligned}$$

$Ax = b$ ← equality constraints

$x \geq 0$ ← non-negative variables

If an LP is not in SEF, then we convert it into an "equivalent" LP in SEF. We can solve the original LP by solving the new LP in SEF using simplex. (If one is infeasible / unbounded, then the other is also infeasible / unbounded. If one has an optimal solution, then we can use it to construct an optimal solution of the other.)

Example:

$$\begin{aligned} \min\ & (3\ \ 7\ \ -2)\,x \\ \text{s.t. } & \begin{pmatrix} 1 & 2 & 3 \\ -4 & 5 & -6 \end{pmatrix} x \begin{matrix} \geq \\ = \end{matrix} \begin{pmatrix} 7 \\ 8 \end{pmatrix} \\ & x_1 \geq 0,\ x_2 \geq 0,\ x_3 \text{ free} \end{aligned}$$

subtract slack var

x_3 free → $x_3^+ - x_3^-$

$$\longrightarrow \quad \begin{aligned} \max\ & (-3\ \ -7\ \ 2\ \ -2\ \ 0)\,x \\ \text{s.t. } & \begin{pmatrix} 1 & 2 & 3 & -3 & -1 \\ -4 & 5 & -6 & 6 & 0 \end{pmatrix} x = \begin{pmatrix} 7 \\ 8 \end{pmatrix} \\ & x_1, x_2, x_3^+, x_3^-, s \geq 0 \end{aligned}$$

How simplex works

Example:

$$\begin{aligned} \max\ & (3\ \ -1\ \ 4\ \ 0\ \ 0)\,x \\ \text{s.t. } & \begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 3 & -1 & 3 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 7 \end{pmatrix} \quad (*) \\ & x \geq 0 \end{aligned}$$

I

Easy feasible solution: $(0\ \ 0\ \ 0\ \ 2\ \ 7)^T$

(Helped by the identity. "basis")

Objective value: 0.

We can raise the objective value by increasing x_1 (positive coefficient in objective)

Say raise x_1 to t, adjust x_4, x_5 (basis), keep feasibility.

$\Rightarrow x_1 = t,\quad x_2 = 0,\quad x_3 = 0,\quad x_4 = 2 - t,\quad x_5 = 7 - 3t$

$(*)\ \ \overset{t}{x_1} - \cancel{2\overset{0}{x_2}} - \cancel{\overset{0}{x_3}} + x_4 = 2$

To keep feasibility, we need $t \geq 0$, $2-t \geq 0$, $7-3t \geq 0$.

$\Rightarrow t \geq 0$, $t \leq 2$, $t \leq \frac{2}{3}$. So the largest t that satisfies all is $t=2$.

New solution $(2\ 0\ 0\ 0\ 1)^T$, new objective $3 \cdot 2 = 6$.

Changing x_4, x_5 does not affect obj since they have coeff 0.

Want: Columns 1,5 become identity (new basis). Coeff of x_1, x_5 in obj are 0.

Basis

If there are m constraints, then any m independent columns form a basis. (They can be turned into I by multiplying by the inverse.) [$B=\{4,5\}$ at the start of the example.]

Basic solution: A solution to $Ax=b$ where all non-basic variables are 0's. For each basis, there is exactly one corresponding basic solution. [$(0\ 0\ 0\ 2\ 7)^T$ in the example.]

Note: a basic variable can still be 0.

Basic feasible solution (BFS): A basic solution that is non-negative.

Simplex needs BFS.

Canonical form

To continue with simplex, we want the LP in canonical form for our basis:

- Identity in the columns of the basis.
- Coeff 0 for the basic variables in the objective.

$$\begin{aligned} \max\ & (3\ \ {-1}\ \ 4\ \ 0\ \ 0)x \\ \text{s.t.}\ & \begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 3 & -1 & 3 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 2 \\ 7 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Canonical form for basis $\{4,5\}$

BFS $(0\ 0\ 0\ 2\ 7)^T$

Want:
$$\begin{aligned} \max\ & (0\ \ ?\ \ ?\ \ ?\ \ 0)x \\ \text{s.t.}\ & \begin{pmatrix} 1 & ? & ? & ? & 0 \\ 0 & ? & ? & ? & 1 \end{pmatrix} x = \begin{pmatrix} ? \\ ? \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

New solution $(2\ 0\ 0\ 0\ 1)^T$

New basis $\{1,5\}$.

To get I for columns 1, 5, we multiply by the inverse of $\begin{pmatrix}1&0\\3&1\end{pmatrix} \leftarrow A_B \quad B=\{1,5\}.$

$\begin{pmatrix}1&0\\3&1\end{pmatrix}^{-1} = \begin{pmatrix}1&0\\-3&1\end{pmatrix}$. Then

$$\begin{pmatrix}1&0\\-3&1\end{pmatrix}\begin{pmatrix}1&2&-1&1&0\\3&-1&3&0&1\end{pmatrix}x = \begin{pmatrix}1&0\\-3&1\end{pmatrix}\begin{pmatrix}2\\7\end{pmatrix} \Rightarrow \begin{pmatrix}1&2&-1&1&0\\0&-7&6&-3&1\end{pmatrix}x = \begin{pmatrix}2\\1\end{pmatrix}.$$

Objective function: Currently $3x_1 - x_2 + 4x_3$. Want to eliminate $x_1 \to 0$.

First row of new constraints: $x_1 + 2x_2 - x_3 + x_4 = 2 \Rightarrow x_1 = 2 - 2x_2 + x_3 - x_4$.

$$3x_1 - x_2 + 4x_3 = 3(2 - 2x_2 + x_3 - x_4) - x_2 + 4x_3 = 6 - 7x_2 + 7x_3 - 3x_4.$$

New canonical form:

$$\begin{aligned} \max\ & (0\ \ -7\ \ 7\ \ -3\ \ 0)x + 6 \\ \text{s.t.}\ & \begin{pmatrix}1&2&-1&1&0\\0&-7&6&-3&1\end{pmatrix}x = \begin{pmatrix}2\\1\end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Basis $\{1,5\}$

BFS $(2\ 0\ 0\ 0\ 1)^T$.

General formula for the objective: $(c^T - y^T A)x + y^T b$ where $y = A_B^{-T} c_B$.

↑ certificate at the end of simplex